WORKSHEET

Chapter 9: Reliability

1 An electronic component has a constant failure rate of 0.04 per hour.

a Calculate the component's reliability at 100 hours.

b If 80 components are tested, calculate the number expected to be in a failed state after 30 hours.

2 A switch has a constant failure rate of 0.3 per hour.

a Calculate the switch's reliability at 100 hours.

b If 300 switches are tested, calculate the number expected to be in a failed state after 10 hours.

3 The length of the life in years, t, of an item is exponentially distributed with $\lambda = 0.5$ years.

a Calculate the probability that the item will last more than a year.

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b When will the reliability of the item be 5% (i.e. there is a 5% chance it still works)?

4 There are 100 of the same item described in Question **3** operating in a manufacturing facility. How long will it take until only 90 of them are still working?

5 Studies of a particular type of machine shows that the time the machine operates before breaking down is exponentially distributed with a mean of 10 hours.

a Determine the failure rate and reliability of the machine.

b Find the probability that the machine operates for at least 12 hours before breaking down.

6 The lifespan in hundreds of hours, t, of a light bulb is exponentially distributed with $\lambda = 0.2$. Calculate the probability that the light bulb will last more than:

a 700 hours

b 900 hours.

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7 A manufacturer has determined that a particular model of its blenders has an expected life that is exponentially distributed with a mean of 4 years. Find the probability that one of these blenders will have a life that ends:

a after the initial 4 years of service

b before 4 years of service are completed.

8 A resistor has an expected life that is exponentially distributed has a constant failure rate (λ) of 0.04 per hour.

a What is the resistor's reliability at 100 hours?

b If 100 resistors are tested, how many would be expected to be in a failed state after 25 hours?

9 A component has an expected life that is exponentially distributed has a constant failure rate (λ) of 2.5×10^{-5} per hour.

a Determine the probability that the component survives a period of 2 months (60 days) without failure.

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b Determine the mean time to failure (MTTF) of the component.

c Determine the probability that the component survives its MTTF.

10 A machine has an expected life that is exponentially distributed with a constant failure rate (λ). The probability that the machine will survive 100 hours without failure is 0.5.

a Determine the failure rate.

b Determine the probability that the machine will survive 500 hours without failure.

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Answers

1 **a** 0.0183

b About 56 (55.904)

2 **a** e^{-30}

b About 285 (285.06)

3 **a** 0.6065

b 6 years

4 0.2107 years

5 **a** Failure rate $= 0.1$ hours; Reliability $= e^{-0.1t}$

b 0.3012

6 **a** 0.2466

b 0.1653

7 **a** 0.3679

b 0.6321

8 **a** 0.018 32

b 64 (63.2)

9 **a** 0.9646

b 40 000 h

c 0.3679

10 **a** 0.006 931

b 0.031 25

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